Systems —

Systems can be of different types and with varying properties. Not all demarcations are of substance to us. For now we will mainly talk about 2 demarcations —

1 Linear vs non-linear systems

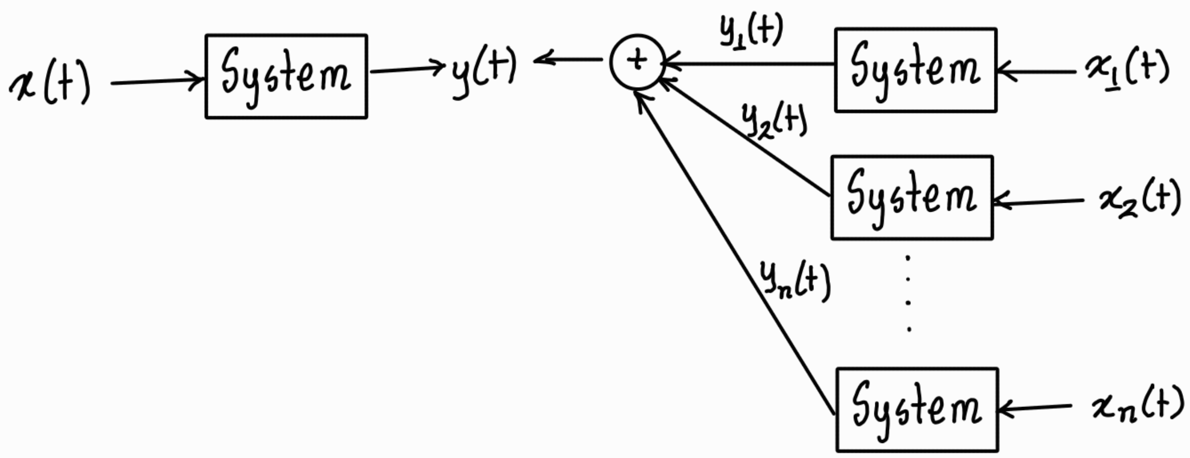
Linear systems satisfy 2 properties —

(i) Additivity —

if $x(t) = x_1(t) + x_2(t) + \dots + x_n(t)$

then $y(t) = y_1(t) + y_2(t) + \dots + y_n(t)$

in terms of schematic,



$x_1(t), x_2(t), \dots, x_n(t)$ can also be the same signal $x(t)$ but time shifted.

(ii) Homogeneity —

if $\text{system}(x(t)) = y(t)$

then $\text{system}(\alpha x(t)) = \alpha y(t)$ given α is a constant

Example —

$$x[n] \longrightarrow \boxed{\text{System}} \longrightarrow y[n] = x[n] + x[n-1] + 2$$ ← check if linear

Let $z[n] = x[n] + x[n-1]$

Then $\text{System}(z[n]) = z[n] + z[n-1] + 2$

$$= x[n] + x[n-1] + x[n-1] + x[n-2] + 2$$

$$= x[n] + 2x[n-1] + x[n-2] + 2$$

and $\text{System}(x[n]) + \text{System}(x[n-1]) = x[n] + x[n-1] + 2 + x[n-1] + x[n-2] + 2$

$$= x[n] + 2x[n-1] + x[n-2] + 4 \neq \text{System}(z[n])$$

Since the system breaks additivity, its a non-linear system.

(2) Time variant vs time invariant systems

If a system doesn't change with time its a time invariant system.

So what this means in simple term is —

if $\text{System}(x(t)) = y(t)$ and $\text{System}(x(t-t_0)) = y(t-t_0)$ $\forall t_0$, then the system is time invariant.

So counter example would be $\text{System}(x(t)) = y_0(t)$ & $\text{System}(x(t-t_0)) = y_{t_0}(t-t_0)$ $\exists t_0$.

Going forward all our systems will be causal LTI (linear time invariant) system.

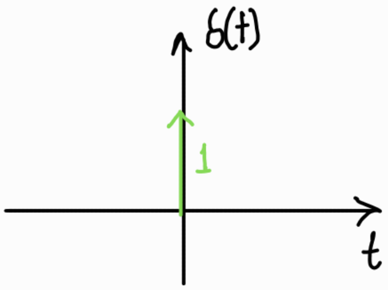
We will develop convolution using LTI properties. But before that we need to learn about a special signal — unit impulse function.

Unit impulse function —

CT definition —

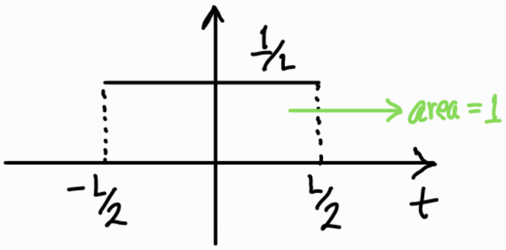
$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0 & \text{elsewhere} \end{cases}$$

property — $\displaystyle\int_{-\infty}^{\infty} \delta(t)\, dt = 1$ ← not mathematically derived but defined

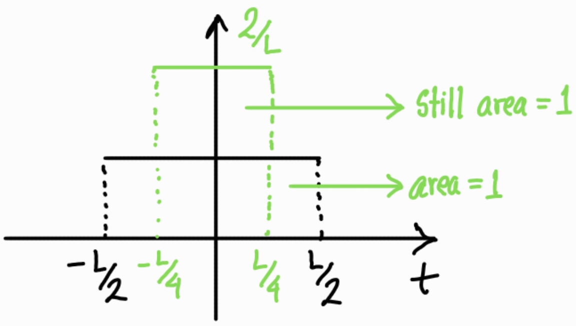


always drawn like this

It was defined as a function with unit area compressed to a single point. You can graphically derive it. Lets start with a function of a box with area 1 —



Now compress the t span



As $\lim_{L \to 0}$ you get the unit impulse function

* You can move the unit impulse function by time shifting —

$$\delta(t-\tau) = \begin{cases} \infty, & t=\tau \\ 0 & \text{otherwise} \end{cases}$$

even when time shifted, $\int_{-\infty}^{\infty} \delta(t-\tau) = 1$

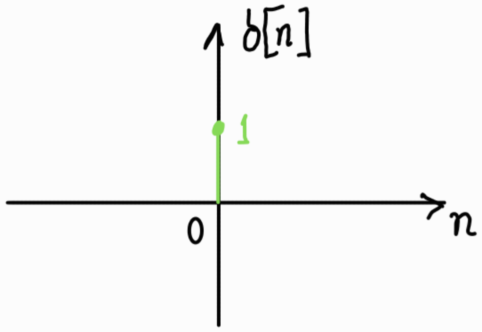
* Multiplying any signal $x(t)$ with a time shifted $\delta(t-\tau)$ will give you $x(\tau)\cdot\delta(t-\tau)$ as $\forall t : t \neq \tau,\ x(t)\cdot\delta(t-\tau) = 0.$

the term persists as value at $t=\tau$ is ∞
We will integrate later to get rid of it.

DT definition —

The DT impulse function is more natural —

$$\delta[n] = \begin{cases} 1 & \text{for } n=0 \\ 0 & \text{otherwise} \end{cases}$$



$$\sum_{n=-\infty}^{\infty} \delta[n] = 1$$

* Time shifted version — $\delta[n-k] = \begin{cases} 1 & \text{for } n=k \\ 0 & \text{otherwise} \end{cases}$

$$\sum_{n=-\infty}^{\infty} \delta[n-k] = 1$$

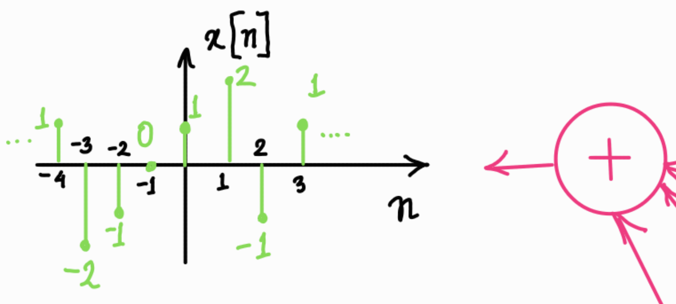
simpler compared to CT as $\delta[n-k]=1$ @ $n=k$

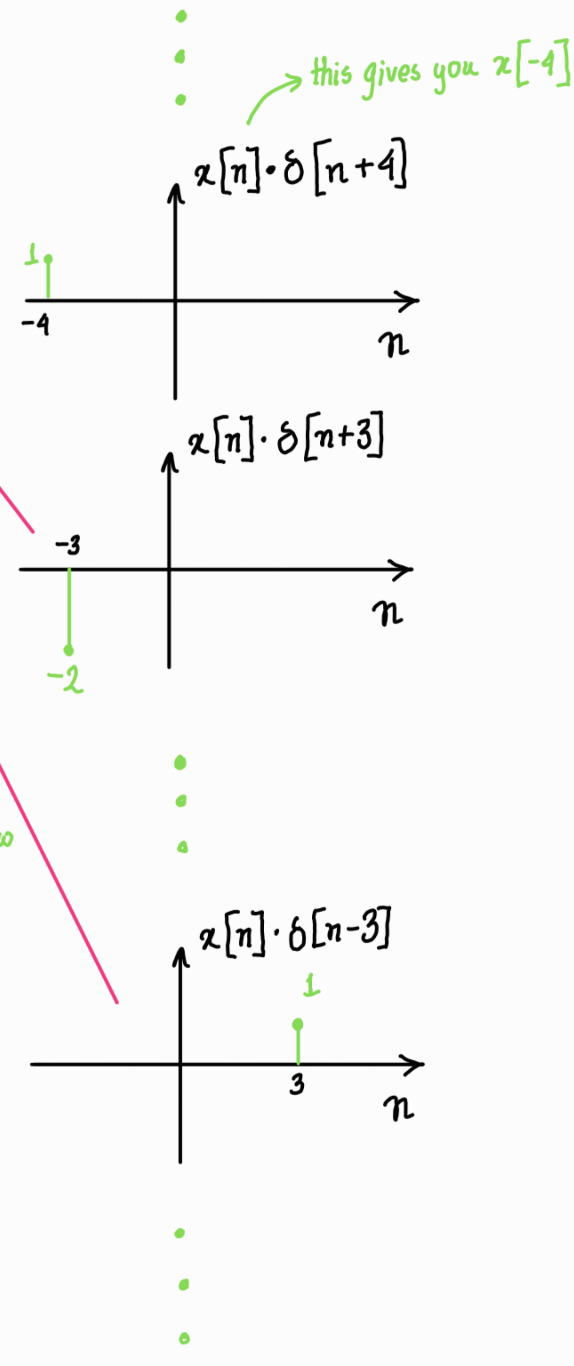
* $x[n]\cdot\delta[n-k] = x[k]\cdot\delta[n-k] = x[k]$

Why go through all the trouble?

We will reformulate our signals using unit impulse signals. Lets start with DT signals —

A random DT signal can be broken down as pulses. For example,





So, $x[n] = \ldots + x[n]\cdot\delta[n+4]$
$+\, x[n]\cdot\delta[n+3]$
$+ \ldots\ldots + x[n]\cdot\delta[n-3]$
$+ \ldots + x[n]\cdot\delta[n-k] + \ldots$

$$= \sum_{k=-\infty}^{\infty} x[n]\cdot\delta[n-k]$$

$$= \sum_{k=-\infty}^{\infty} \underbrace{x[k]}_{\text{this is a constant now}}\cdot\underbrace{\delta[n-k]}_{\text{the impulse is the signal now}}$$

the impulse is the signal now

this is a constant now
assuming we have all the values of $x[n]$

This reformulation is called the sifting property.

How does this reformulation help?

Lets say we have a causal LTI system —

$$x[n] \longrightarrow \boxed{\text{System}} \longrightarrow y[n]$$

then $y[n] = \text{System}(x[n]) = \text{System}\left(\sum_{k=-\infty}^{\infty} x[k]\cdot\delta[n-k]\right)$

Since the system is LTI and obeys additivity —

$$\text{System}\left(\sum_{k=-\infty}^{\infty} x[k]\cdot\delta[n-k]\right) = \sum_{k=-\infty}^{\infty} \text{System}\left(x[k]\cdot\delta[n-k]\right)$$

As per homogeneity principle, as $x[k]$ is a constant —

$$\sum_{k=-\infty}^{\infty} \text{System}\left(x[k]\cdot\delta[n-k]\right) = \sum_{k=-\infty}^{\infty} x[k]\cdot\text{System}\left(\delta[n-k]\right)$$

Let, $\text{System}(\delta[n]) = h[n]$ ⟵ a signal, formally called unit impulse response

Now, as the system is time invariant, $\text{System}(\delta[n-k]) = h[n-k]$

So, we have —

$$\sum_{k=-\infty}^{\infty} x[k]\cdot\text{System}\left(\delta[n-k]\right) = \sum_{k=-\infty}^{\infty} x[k]\cdot h[n-k]$$ ⟵ this is the convolution operation in DT domain

$$= x[n] * \boxed{h[n]}$$ ⟶ contains all the system info but as a signal

↑ convolution operator

So, I hope the build up did pay off. We are yet to understand how convolution physically works. We will see that in the next lecture. In rest of the lecture we will derive CT convolution.

First, we will derive the sifted reformulation of $x(t)$. If we want to pick a random point, we can just multiply $x(t)$ by $\delta(t-\tau)$. But the problem is it will give us $x(\tau)\cdot\delta(t-\tau)$. $\delta(t-\tau)$ gives ∞ at $t=\tau$, and so we cannot directly get $x(\tau)$ like we did in case of DT. But I mentioned that it will get sorted out using an integration operation.

Lets ignore the problem and instead focus on getting the whole signal for now.

Assuming $x(\tau)\cdot\delta(t-\tau)$ gives info for one point in a continuous signal at $t=\tau$. We just have to integrate wrt τ to get the whole signal info —

Sifted info, $\hat{x}(t) = \int_{-\infty}^{\infty} x(\tau)\cdot\delta(t-\tau)\,d\tau$

This actually solves our previous problem as

$$\hat{x}(t) = \int_{-\infty}^{\infty} x(\tau)\cdot\delta(t-\tau)\,d\tau$$

$$= \int_{-\infty}^{\infty} x(t)\cdot\delta(t-\tau)\,d\tau$$

$$= x(t)\int_{-\infty}^{\infty} \delta(t-\tau)\,d\tau$$

$$= x(t)$$

So, $x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$ ⟵ sifting property of CT impulse function

So, when passed through any system following LTI formulation —

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\cdot h(t-\tau)\,d\tau = x(t) * h(t)$$

Derivation is same as before.